

# Database System Principle - Relational Model

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# OBJECTIVES

- Structure of Relational Databases
- Database Schema
- Keys
- Schema Diagrams
- Relational Query Languages
- Relational Operations

# STRUCTURE OF RELATIONAL DATABASES

- A relational database consists of a collection of **tables**, each of which is assigned a unique name

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> |
|-----------|-------------|------------------|---------------|
| 10101     | Srinivasan  | Comp. Sci.       | 65000         |
| 12121     | Wu          | Finance          | 90000         |
| 15151     | Mozart      | Music            | 40000         |
| 22222     | Einstein    | Physics          | 95000         |
| 32343     | El Said     | History          | 60000         |
| 33456     | Gold        | Physics          | 87000         |
| 45565     | Katz        | Comp. Sci.       | 75000         |
| 58583     | Califieri   | History          | 62000         |
| 76543     | Singh       | Finance          | 80000         |
| 76766     | Crick       | Biology          | 72000         |
| 83821     | Brandt      | Comp. Sci.       | 92000         |
| 98345     | Kim         | Elec. Eng.       | 80000         |

attributes  
(or columns)

tuples  
(or rows)

Table: instructor



# STRUCTURE OF RELATIONAL DATABASES

- Relational database
  - consists of a collection of tables, each of which is assigned a unique name
  - Table: **Relation**
  - Row: **Tuple**
  - Column: **attribute**
    - Domain(域)\*: a set of permitted values for one attribute

# STRUCTURE OF RELATIONAL DATABASES

## - ATTRIBUTE TYPES

- Domain
  - a set of allowed values for one attribute
- Attribute values are (normally) required to be **atomic**; that is, **indivisible**(不可再分)
- The special value **null** is a member of every domain. Indicated that the value is “unknown”
- The null value causes complications in the definition of many operations

# RELATION, RELATION SCHEMA AND INSTANCE

## - RELATION

- $A_1, A_2, \dots, A_n$  are *attributes*
- given sets  $D_1, D_2, \dots, D_n$ , a **relation**  $r$  is a subset of
$$D_1 \times D_2 \times \dots \times D_n$$
- Thus, a **relation** is a set of  $n$ -tuples  $(a_1, a_2, \dots, a_n)$  where each  $a_i \in D_i$
- An element  $t$  of  $r$  is a **tuple**, represented by a *row* in a table

# RELATION, RELATION SCHEMA AND INSTANCE

## - RELATION SCHEMA 关系模式

- $A_1, A_2, \dots, A_n$  are *attributes*
- $R = (A_1, A_2, \dots, A_n)$  is a *relation schema* (关系模式)

Example: *instructor* = (*ID*, *name*, *dept\_name*, *salary*)



# RELATION SCHEMA: CASES

- student ( ID , name, dept\_name, tot\_cred)
- advisor (s\_id, i\_id)
- takes ( ID , course\_id, sec\_id, semester, year, grade)
- classroom (building, room\_number, capacity)
- time\_slot (time\_slot\_id, day, start\_time, end\_time)



# RELATION, RELATION SCHEMA AND INSTANCE

## - RELATION INSTANCE

- Relation Instance 关系实例
  - A specific instance of a relation, i.e., containing a specific set of rows
- Question
  - What is the difference between relation schema and relation instance?

# RELATION INSTANCE: CASE

- Course table

| <i>course_id</i> | <i>title</i>               | <i>dept_name</i> | <i>credits</i> |
|------------------|----------------------------|------------------|----------------|
| BIO-101          | Intro. to Biology          | Biology          | 4              |
| BIO-301          | Genetics                   | Biology          | 4              |
| BIO-399          | Computational Biology      | Biology          | 3              |
| CS-101           | Intro. to Computer Science | Comp. Sci.       | 4              |
| CS-190           | Game Design                | Comp. Sci.       | 4              |
| CS-315           | Robotics                   | Comp. Sci.       | 3              |
| CS-319           | Image Processing           | Comp. Sci.       | 3              |
| CS-347           | Database System Concepts   | Comp. Sci.       | 3              |
| EE-181           | Intro. to Digital Systems  | Elec. Eng.       | 3              |
| FIN-201          | Investment Banking         | Finance          | 3              |
| HIS-351          | World History              | History          | 3              |
| MU-199           | Music Video Production     | Music            | 3              |
| PHY-101          | Physical Principles        | Physics          | 4              |

# RELATION INSTANCE: CASE I

- Prereq table

| <i>course_id</i> | <i>prereq_id</i> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| BIO-399          | BIO-101          |
| CS-190           | CS-101           |
| CS-315           | CS-101           |
| CS-319           | CS-101           |
| CS-347           | CS-101           |
| EE-181           | PHY-101          |



# RELATION INSTANCE: CASE II

- section table

| <i>course_id</i> | <i>sec_id</i> | <i>semester</i> | <i>year</i> | <i>building</i> | <i>room_number</i> | <i>time_slot_id</i> |
|------------------|---------------|-----------------|-------------|-----------------|--------------------|---------------------|
| BIO-101          | 1             | Summer          | 2017        | Painter         | 514                | B                   |
| BIO-301          | 1             | Summer          | 2018        | Painter         | 514                | A                   |
| CS-101           | 1             | Fall            | 2017        | Packard         | 101                | H                   |
| CS-101           | 1             | Spring          | 2018        | Packard         | 101                | F                   |
| CS-190           | 1             | Spring          | 2017        | Taylor          | 3128               | E                   |
| CS-190           | 2             | Spring          | 2017        | Taylor          | 3128               | A                   |
| CS-315           | 1             | Spring          | 2018        | Watson          | 120                | D                   |
| CS-319           | 1             | Spring          | 2018        | Watson          | 100                | B                   |
| CS-319           | 2             | Spring          | 2018        | Taylor          | 3128               | C                   |
| CS-347           | 1             | Fall            | 2017        | Taylor          | 3128               | A                   |
| EE-181           | 1             | Spring          | 2017        | Taylor          | 3128               | C                   |
| FIN-201          | 1             | Spring          | 2018        | Packard         | 101                | B                   |
| HIS-351          | 1             | Spring          | 2018        | Painter         | 514                | C                   |
| MU-199           | 1             | Spring          | 2018        | Packard         | 101                | D                   |
| PHY-101          | 1             | Fall            | 2017        | Watson          | 100                | A                   |

# RELATIONS ARE UNORDERED

- Order of tuples is irrelevant (tuples may be stored in an arbitrary order)
- Example: instructor relation with unordered tuples

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> |
|-----------|-------------|------------------|---------------|
| 22222     | Einstein    | Physics          | 95000         |
| 12121     | Wu          | Finance          | 90000         |
| 32343     | El Said     | History          | 60000         |
| 45565     | Katz        | Comp. Sci.       | 75000         |
| 98345     | Kim         | Elec. Eng.       | 80000         |
| 76766     | Crick       | Biology          | 72000         |
| 10101     | Srinivasan  | Comp. Sci.       | 65000         |
| 58583     | Califieri   | History          | 62000         |
| 83821     | Brandt      | Comp. Sci.       | 92000         |
| 15151     | Mozart      | Music            | 40000         |
| 33456     | Gold        | Physics          | 87000         |
| 76543     | Singh       | Finance          | 80000         |

# KEYS码/键

- Let  $K \subseteq R$
- $K$  is a **superkey 超码** of  $R$  if values for  $K$  are sufficient to identify a unique tuple of each possible relation  $r(R)$ 
  - Example:  $\{ID\}$  and  $\{ID, name\}$  are both superkeys of *instructor*.
- Superkey  $K$  is a **candidate key 候选码** if  $K$  is minimal  
Example:  $\{ID\}$  is a candidate key for *Instructor*
- One of the candidate keys is selected to be the **primary key 主码**.\*



# WHICH ONE IS PRIMARY KEY?

- If two candidate keys
  - {ID}
  - {name, dept\_name}

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> |
|-----------|-------------|------------------|---------------|
| 10101     | Srinivasan  | Comp. Sci.       | 65000         |
| 12121     | Wu          | Finance          | 90000         |
| 15151     | Mozart      | Music            | 40000         |
| 22222     | Einstein    | Physics          | 95000         |
| 32343     | El Said     | History          | 60000         |
| 33456     | Gold        | Physics          | 87000         |
| 45565     | Katz        | Comp. Sci.       | 75000         |
| 58583     | Califieri   | History          | 62000         |
| 76543     | Singh       | Finance          | 80000         |
| 76766     | Crick       | Biology          | 72000         |
| 83821     | Brandt      | Comp. Sci.       | 92000         |
| 98345     | Kim         | Elec. Eng.       | 80000         |

Table: instructor



# KEYS

## - PRIMARY KEY

- Primary keys

- It denote **a candidate** key that is chosen by the database designer as the principal means of identifying tuples within a relation
- *It must be chosen with care*
  - Example: the name of a person is obviously not sufficient

# KEYS

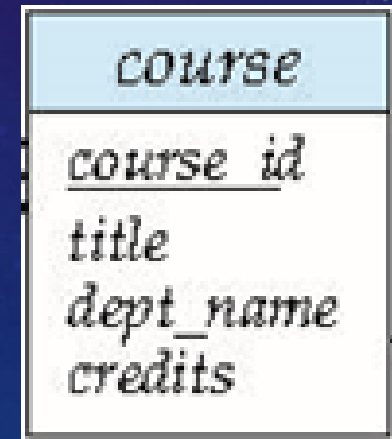
## - FOREIGN KEY

- **Foreign key**外码
  - the key constraint: Value in one relation must appear in another
  - **Referencing** relation参照关系
  - **Referenced** relation被参照关系
  - Example – *dept\_name* in *instructor* is a foreign key from *instructor* referencing *department*
- *Referential integrity constraint*参照完整性约束

# SCHEMA DIAGRAM

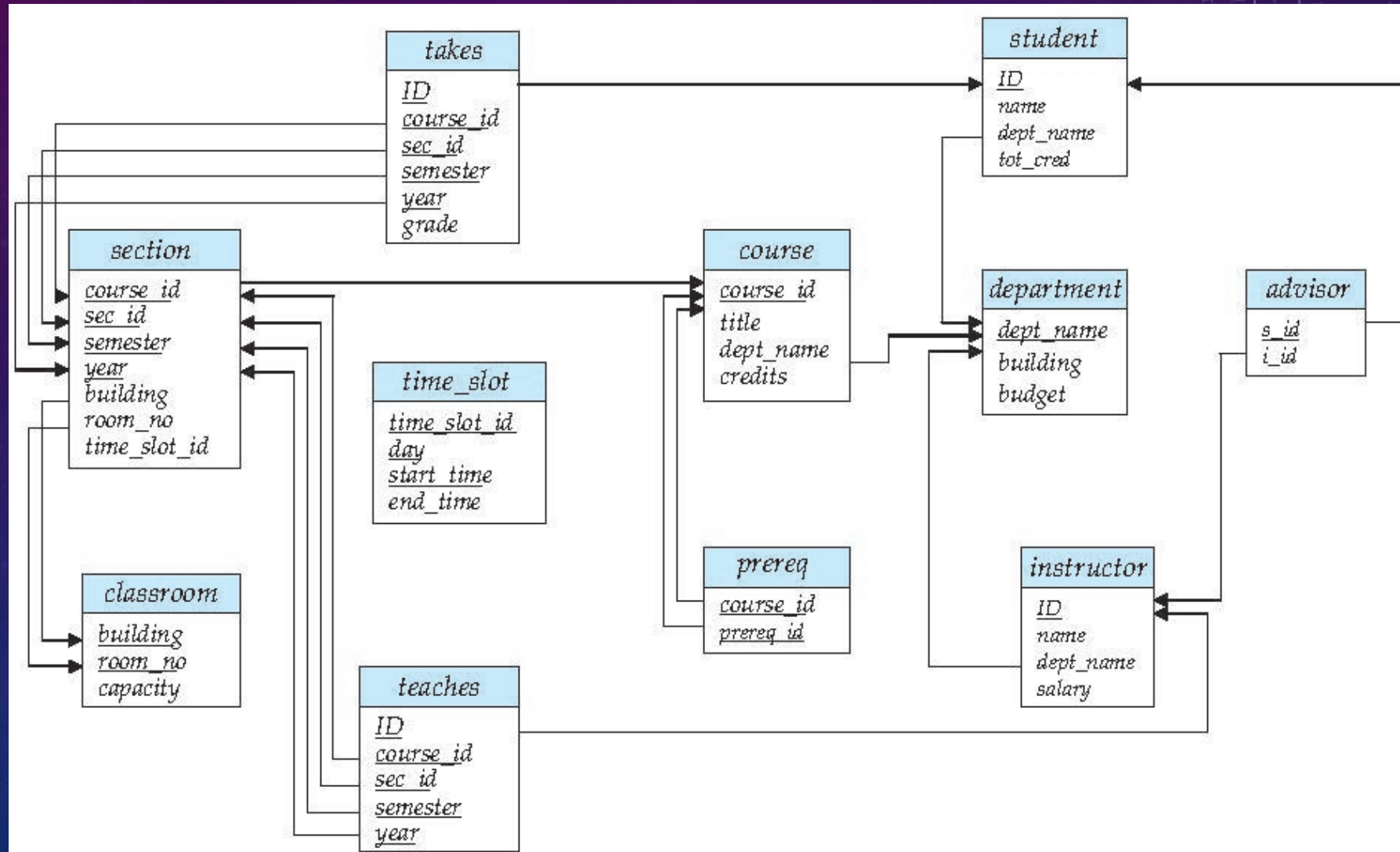
- **Schema diagram** 模式图

- A database schema, along with primary key and foreign key dependencies, can be depicted by schema diagrams
- Each relation appears as a **box**
  - with the relation name at the top in blue
  - And the attributes listed inside the box
  - Primary key attributes are shown **underlined**
  - Foreign key dependencies appear as **arrows** from the foreign key attributes of the referencing relation to the primary key of the referenced relation





# SCHEMA DIAGRAM: CASE - UNIVERSITY DATABASE





# SCHEMA OF THE UNIVERSITY DATABASE

```
classroom(building, room_number, capacity)
department(dept_name, building, budget)
course(course_id, title, dept_name, credits)
instructor(ID, name, dept_name, salary)
section(course_id, sec_id, semester, year, building, room_number, time_slot_id)
teaches(ID, course_id, sec_id, semester, year)
student(ID, name, dept_name, tot_cred)
takes(ID, course_id, sec_id, semester, year, grade)
advisor(s_ID, i_ID)
time_slot(time_slot_id, day, start_time, end_time)
prereq(course_id, prereq_id)
```

# RELATIONAL QUERY LANGUAGES

- Query language
  - A language in which a user requests information from the database
- Categories
  - Imperative language 命令式语言: the user instructs the system to perform a specific sequence of operations on the database to compute the desired result
  - functional query language 函数式查询语言: the computation is expressed as the evaluation of functions that may operate on data in the database or on the results of other functions; functions are side-effect free, and they do not update the program state
  - Non procedural language 非过程式语言: declarative

# RELATIONAL QUERY LANGUAGES

- “Pure” languages:
  - 1. Relational algebra关系代数
    - The relational algebra consists of a set of operations that take one or two relations as input and produce a new relation as their result.
    - (belong to procedural language)
  - 2. Tuple relational calculus元组关系演算
  - 3. Domain relational calculus域关系演算
- The above 3 pure languages are **equivalent** in computing power
  - Relational operations: 6 basic operations\*

# RELATIONAL OPERATIONS

- Relational Operations
  - Select specific tuples from a single relation
  - Select certain attributed from a relation
  - Combine two relations by merging pairs of tuples
  - A set union of two similarly structured tables



# RELATIONAL OPERATIONS

## - SELECTION: SELECTION OF ROWS (TUPLES)

□ Relation r

| A        | B        | C  | D  |
|----------|----------|----|----|
| $\alpha$ | $\alpha$ | 1  | 7  |
| $\alpha$ | $\beta$  | 5  | 7  |
| $\beta$  | $\beta$  | 12 | 3  |
| $\beta$  | $\beta$  | 23 | 10 |

■  $\sigma_{A=B \wedge D > 5}(r)$

| A        | B        | C  | D  |
|----------|----------|----|----|
| $\alpha$ | $\alpha$ | 1  | 7  |
| $\beta$  | $\beta$  | 23 | 10 |

# RELATIONAL OPERATIONS

## - SELECTION: SELECTION OF COLUMNS (ATTRIBUTES)

- Relation  $r$ :

| A        | B  | C |
|----------|----|---|
| $\alpha$ | 10 | 1 |
| $\alpha$ | 20 | 1 |
| $\beta$  | 30 | 1 |
| $\beta$  | 40 | 2 |

$$\Pi_{A,C}(r)$$

| A        | C |
|----------|---|
| $\alpha$ | 1 |
| $\alpha$ | 1 |
| $\beta$  | 1 |
| $\beta$  | 2 |

=

| A        | C |
|----------|---|
| $\alpha$ | 1 |
| $\beta$  | 1 |
| $\beta$  | 2 |

OK

# RELATIONAL OPERATIONS

## - COMPOSITION OF RELATIONAL OPERATIONS

- Instructor table

$\Pi_{name} (\sigma_{dept\_name = \text{"Physics"}} (instructor))$

| <i>ID</i> | <i>name</i> | <i>salary</i> |
|-----------|-------------|---------------|
| 10101     | Srinivasan  | 65000         |
| 12121     | Wu          | 90000         |
| 15151     | Mozart      | 40000         |
| 22222     | Einstein    | 95000         |
| 32343     | El Said     | 60000         |
| 33456     | Gold        | 87000         |
| 45565     | Katz        | 75000         |
| 58583     | Califieri   | 62000         |
| 76543     | Singh       | 80000         |
| 76766     | Crick       | 72000         |
| 83821     | Brandt      | 92000         |
| 98345     | Kim         | 80000         |

# RELATIONAL OPERATIONS

## - UNION OF TWO RELATIONS

- Relations  $r, s$ :

| A        | B |
|----------|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$  | 1 |

$r$

| A        | B |
|----------|---|
| $\alpha$ | 2 |
| $\beta$  | 3 |

$s$

| A        | B |
|----------|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$  | 1 |
| $\beta$  | 3 |

□  $r \cup s$ :



# RELATIONAL OPERATIONS

## - SET DIFFERENCE OF TWO RELATIONS

- Relations  $r, s$ :

| $A$      | $B$ |
|----------|-----|
| $\alpha$ | 1   |
| $\alpha$ | 2   |
| $\beta$  | 1   |

$r$

| $A$      | $B$ |
|----------|-----|
| $\alpha$ | 2   |
| $\beta$  | 3   |

$s$

| $A$      | $B$ |
|----------|-----|
| $\alpha$ | 1   |
| $\beta$  | 1   |

□  $r - s$ :

# RELATIONAL OPERATIONS

## - SET INTERSECTION OF TWO RELATIONS

- Relation  $r, s$ :

- $r \cap s$

Note:  $r \cap s = r - (r - s)$

| A        | B |
|----------|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$  | 1 |

$r$

| A        | B |
|----------|---|
| $\alpha$ | 2 |
| $\beta$  | 3 |

$s$

| A        | B |
|----------|---|
| $\alpha$ | 2 |

# RELATIONAL OPERATIONS

## - JOIN: CARTESIAN-PRODUCT笛卡尔积

□ Relations  $r, s$ :

| A        | B | C        | D  | E |
|----------|---|----------|----|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\beta$  | 2 | $\beta$  | 10 | a |
|          |   | $\beta$  | 20 | b |
|          |   | $\gamma$ | 10 | b |

$r$   $s$

□  $r \times s$ :

| A        | B | C        | D  | E |
|----------|---|----------|----|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$  | 10 | a |
| $\alpha$ | 1 | $\beta$  | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$  | 2 | $\alpha$ | 10 | a |
| $\beta$  | 2 | $\beta$  | 10 | a |
| $\beta$  | 2 | $\beta$  | 20 | b |
| $\beta$  | 2 | $\gamma$ | 10 | b |

Note: Regardless of whether or not they have the same values on common attributed



# RELATIONAL OPERATIONS: CARTESIAN-PRODUCT

## - CASE II

- the Cartesian product  
instructor  $\times$  teaches

| <i>instructor.ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> | <i>teaches.ID</i> | <i>course_id</i> | <i>sec_id</i> | <i>semester</i> | <i>year</i> |
|----------------------|-------------|------------------|---------------|-------------------|------------------|---------------|-----------------|-------------|
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-101           | 1             | Fall            | 2017        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-315           | 1             | Spring          | 2018        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-347           | 1             | Fall            | 2017        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 12121             | FIN-201          | 1             | Spring          | 2018        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 15151             | MU-199           | 1             | Spring          | 2018        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 22222             | PHY-101          | 1             | Fall            | 2017        |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| 12121                | Wu          | Finance          | 90000         | 10101             | CS-101           | 1             | Fall            | 2017        |
| 12121                | Wu          | Finance          | 90000         | 10101             | CS-315           | 1             | Spring          | 2018        |
| 12121                | Wu          | Finance          | 90000         | 10101             | CS-347           | 1             | Fall            | 2017        |
| 12121                | Wu          | Finance          | 90000         | 12121             | FIN-201          | 1             | Spring          | 2018        |
| 12121                | Wu          | Finance          | 90000         | 15151             | MU-199           | 1             | Spring          | 2018        |
| 12121                | Wu          | Finance          | 90000         | 22222             | PHY-101          | 1             | Fall            | 2017        |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| 15151                | Mozart      | Music            | 40000         | 10101             | CS-101           | 1             | Fall            | 2017        |
| 15151                | Mozart      | Music            | 40000         | 10101             | CS-315           | 1             | Spring          | 2018        |
| 15151                | Mozart      | Music            | 40000         | 10101             | CS-347           | 1             | Fall            | 2017        |
| 15151                | Mozart      | Music            | 40000         | 12121             | FIN-201          | 1             | Spring          | 2018        |
| 15151                | Mozart      | Music            | 40000         | 15151             | MU-199           | 1             | Spring          | 2018        |
| 15151                | Mozart      | Music            | 40000         | 22222             | PHY-101          | 1             | Fall            | 2017        |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| 22222                | Einstein    | Physics          | 95000         | 10101             | CS-101           | 1             | Fall            | 2017        |
| 22222                | Einstein    | Physics          | 95000         | 10101             | CS-315           | 1             | Spring          | 2018        |
| 22222                | Einstein    | Physics          | 95000         | 10101             | CS-347           | 1             | Fall            | 2017        |
| 22222                | Einstein    | Physics          | 95000         | 12121             | FIN-201          | 1             | Spring          | 2018        |
| 22222                | Einstein    | Physics          | 95000         | 15151             | MU-199           | 1             | Spring          | 2018        |
| 22222                | Einstein    | Physics          | 95000         | 22222             | PHY-101          | 1             | Fall            | 2017        |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |
| ...                  | ...         | ...              | ...           | ...               | ...              | ...           | ...             | ...         |



CARTESIAN 笛卡尔

I think; therefore I am



# RELATIONAL OPERATIONS

## - RENAMING A TABLE

- Allows us to refer to a relation, (say  $E$ ) by more than one name.

$$\rho_x(E)$$

returns the expression  $E$  under the name  $X$

□ Relations  $r$

| $A$      | $B$ |
|----------|-----|
| $\alpha$ | 1   |
| $\beta$  | 2   |

$r$

□  $r \times \rho_s(r)$

| $r.A$    | $r.B$ | $s.A$    | $s.B$ |
|----------|-------|----------|-------|
| $\alpha$ | 1     | $\alpha$ | 1     |
| $\alpha$ | 1     | $\beta$  | 2     |
| $\beta$  | 2     | $\alpha$ | 1     |
| $\beta$  | 2     | $\beta$  | 2     |

# RELATIONAL OPERATIONS

## - JOIN: NATURAL JOIN 自然连接

- Let  $r$  and  $s$  be relations on schemas  $R$  and  $S$  respectively. Then, the “**natural join**” of relations  $R$  and  $S$  is a relation on schema  $R \cup S$  obtained as follows:
  - Consider each pair of tuples  $t_r$  from  $r$  and  $t_s$  from  $s$ .
  - If  $t_r$  and  $t_s$  have the same value on each of the attributes in  $R \cap S$ , add a tuple  $t$  to the result, **where**
    - $t$  has the same value as  $t_r$  on  $r$
    - $t$  has the same value as  $t_s$  on  $s$



# RELATIONAL OPERATIONS

## - JOIN: NATURAL JOIN -> CASES

- Relations  $r, s$ :

□ Natural Join

□  $r \bowtie s$

$$\Pi_{A, r.B, C, r.D, E}(\sigma_{r.B = s.B \wedge r.D = s.D}(r \times s))$$

| A        | B | C        | D |
|----------|---|----------|---|
| $\alpha$ | 1 | $\alpha$ | a |
| $\beta$  | 2 | $\gamma$ | a |
| $\gamma$ | 4 | $\beta$  | b |
| $\alpha$ | 1 | $\gamma$ | a |
| $\delta$ | 2 | $\beta$  | b |

$r$

| B | D | E          |
|---|---|------------|
| 1 | a | $\alpha$   |
| 3 | a | $\beta$    |
| 1 | a | $\gamma$   |
| 2 | b | $\delta$   |
| 3 | b | $\epsilon$ |

$s$

| A        | B | C        | D | E        |
|----------|---|----------|---|----------|
| $\alpha$ | 1 | $\alpha$ | a | $\alpha$ |
| $\alpha$ | 1 | $\alpha$ | a | $\gamma$ |
| $\alpha$ | 1 | $\gamma$ | a | $\alpha$ |
| $\alpha$ | 1 | $\gamma$ | a | $\gamma$ |
| $\delta$ | 2 | $\beta$  | b | $\delta$ |



# RELATIONAL OPERATIONS

$\sigma_{instructor.ID = teaches.ID}(instructor \times teaches)$

## - JOIN: NATURAL JOIN -> CASES

| <i>instructor.ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> | <i>teaches.ID</i> | <i>course_id</i> | <i>sec_id</i> | <i>semester</i> | <i>year</i> |
|----------------------|-------------|------------------|---------------|-------------------|------------------|---------------|-----------------|-------------|
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-101           | 1             | Fall            | 2017        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-315           | 1             | Spring          | 2018        |
| 10101                | Srinivasan  | Comp. Sci.       | 65000         | 10101             | CS-347           | 1             | Fall            | 2017        |
| 12121                | Wu          | Finance          | 90000         | 12121             | FIN-201          | 1             | Spring          | 2018        |
| 15151                | Mozart      | Music            | 40000         | 15151             | MU-199           | 1             | Spring          | 2018        |
| 22222                | Einstein    | Physics          | 95000         | 22222             | PHY-101          | 1             | Fall            | 2017        |
| 32343                | El Said     | History          | 60000         | 32343             | HIS-351          | 1             | Spring          | 2018        |
| 45565                | Katz        | Comp. Sci.       | 75000         | 45565             | CS-101           | 1             | Spring          | 2018        |
| 45565                | Katz        | Comp. Sci.       | 75000         | 45565             | CS-319           | 1             | Spring          | 2018        |
| 76766                | Crick       | Biology          | 72000         | 76766             | BIO-101          | 1             | Summer          | 2017        |
| 76766                | Crick       | Biology          | 72000         | 76766             | BIO-301          | 1             | Summer          | 2018        |
| 83821                | Brandt      | Comp. Sci.       | 92000         | 83821             | CS-190           | 1             | Spring          | 2017        |
| 83821                | Brandt      | Comp. Sci.       | 92000         | 83821             | CS-190           | 2             | Spring          | 2017        |
| 83821                | Brandt      | Comp. Sci.       | 92000         | 83821             | CS-319           | 2             | Spring          | 2018        |
| 98345                | Kim         | Elec. Eng.       | 80000         | 98345             | EE-181           | 1             | Spring          | 2017        |

# NOTES ABOUT RELATIONAL LANGUAGES

- Each Query **input** is a table (or set of tables)
- Each query **output** is a table.
- All data in the output table appears in one of the input tables
- Relational Algebra is **not** Turing complete
- Can we compute:
  - SUM, AVG, MAX, MIN

# RELATIONAL OPERATIONS

| Symbol (Name)                   | Example of Use                                                                                                                      |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $\sigma$<br>(Selection)         | $\sigma_{\text{salary} \geq 85000}(\text{instructor})$                                                                              |
|                                 | Return rows of the input relation that satisfy the predicate.                                                                       |
| $\Pi$<br>(Projection)           | $\Pi_{ID, salary}(\text{instructor})$                                                                                               |
|                                 | Output specified attributes from all rows of the input relation. Remove duplicate tuples from the output.                           |
| $\bowtie$<br>(Natural join)     | $\text{instructor} \bowtie \text{department}$                                                                                       |
|                                 | Output pairs of rows from the two input relations that have the same value on all attributes that have the same name.               |
| $\times$<br>(Cartesian product) | $\text{instructor} \times \text{department}$                                                                                        |
|                                 | Output all pairs of rows from the two input relations (regardless of whether or not they have the same values on common attributes) |
| $\cup$<br>(Union)               | $\Pi_{name}(\text{instructor}) \cup \Pi_{name}(\text{student})$<br>Output the union of tuples from the two input relations.         |

# EQUIVALENT QUERIES

$$\sigma_{dept\_name = \text{"Physics"}}(instructor \bowtie_{instructor.ID = teaches.ID} teaches)$$
$$(\sigma_{dept\_name = \text{"Physics"}}(instructor)) \bowtie_{instructor.ID = teaches.ID} teaches$$

- **Query optimizers** in database systems typically look at what result an expression computes and find an efficient way of computing that result, rather than following the exact sequence of steps specified in the query.
- The algebraic structure of **relational algebra** makes it easy to find efficient but equivalent alternative expressions



# SUMMARY

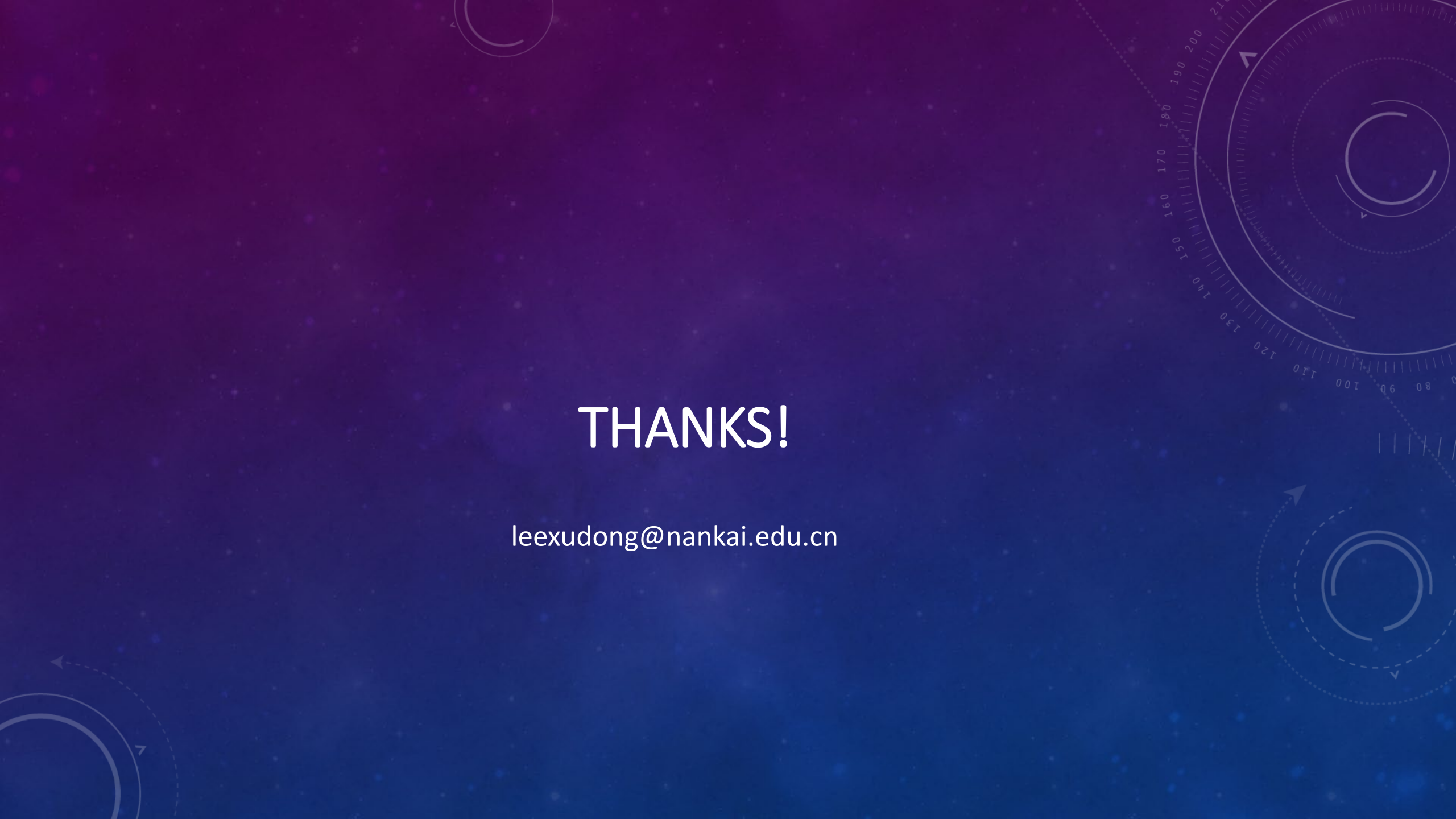
- Structure of Relational Databases
- Database Schema
- Keys
- Schema Diagrams
- Relational Query Languages
- Relational Operations

Q&A?

# QUIZ

- Consider the bank database of the Figure. Give an expression in the relational algebra for each of the following queries:
  - a. Find the name of each branch located in “Chicago”.
  - b. Find the ID of each borrower who has a loan in branch “Downtown”.

```
branch(branch_name, branch_city, assets)  
customer (ID, customer_name, customer_street, customer_city)  
loan (loan_number, branch_name, amount)  
borrower (ID, loan_number)  
account (account_number, branch_name, balance)  
depositor (ID, account_number)
```



# THANKS!

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